中英对照。英文为原文，中文为译文。类名、参数名、函数名、资源路径与网址保持英文，便于在虚幻引擎细节面板中定位。插图、标题和条目顺序与原 PDF 一致。

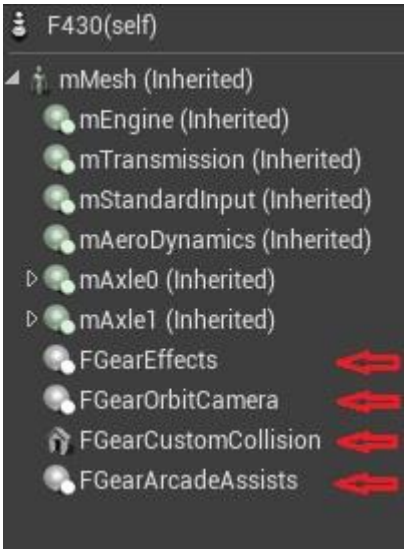


Extras

扩展功能

These components are not actually a part of vehicle physics but complementary tools that were used while making the demo scenes. You can use them for prototyping or learning purposes but they are far from being robust solutions and not recommended for production usage.

这些组件其实不属于载具物理本身，而是制作演示场景时使用的配套工具。可以用来做原型或学习，但远不是稳健的方案，不建议用于正式产品。



UFGearOrbitCamera : A simple camera component that orbits around the vehicle. If the spring parameter is non zero then the yaw rotation is interpolated towards the vehicles yaw. If DriftMode is enabled this time the yaw rotation is interpolated towards the vehicles movement vector. Distance units are in regular unreal units(cm). You can also look back by altering an internal variable, check the sandbox map for a demonstration(press the middle mouse to look back).

UFGearOrbitCamera：一个绕载具环绕的简单摄像机组件。若 spring 参数不为零，偏航旋转会向载具的偏航插值。若启用 DriftMode，则偏航旋转改为向载具的运动向量插值。距离单位是虚幻常规单位（厘米）。也可以通过修改一个内部变量向后看，沙盒地图中有演示（按鼠标中键向后看）。



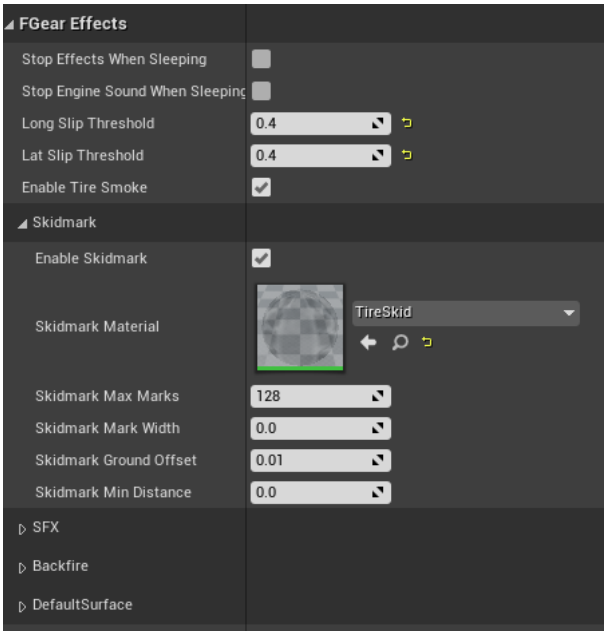
UFGearEffects : This component lets you add some visual and sound effects to your vehicle including engine sound, particle effects and skidmarks. Additionaly if you derive from FGearPhysicalMaterial and assign to road surfaces you can set the surface properties and get physical and visual feedback.

UFGearEffects：该组件可以为载具添加视觉和声音效果，包括发动机声、粒子效果和轮胎印。另外，如果

从 FGearPhysicalMaterial 派生并指定给路面，就可以设置表面属性，并得到物理与视觉反馈。

FGearEffects/Skidmark

FGearEffects / 轮胎印



• Stop Effects When Sleeping : If set, skid sound and particle effects are paused when the vehicles goes to sleep.

Stop Effects When Sleeping（休眠时停止效果）：若启用，载具进入休眠时会暂停打滑声和粒子效果。

• Stop Engine Sound When Sleeping : If set, engine sound is paused when the vehicles goes to sleep.

Stop Engine Sound When Sleeping（休眠时停止发动机声）：若启用，载具进入休眠时会暂停发动机声。

• Long Slip Threshold : If the longitudinal slip is over this value, skidmark, skid sound and particle effects are triggered.

Long Slip Threshold（纵向滑移阈值）：纵向滑移超过该值时，触发轮胎印、打滑声和粒子效果。

• Lat Slip Threshold : If the lateral slip is over this value, skidmark, skid sound and particle effects are triggered.

Lat Slip Threshold（侧向滑移阈值）：侧向滑移超过该值时，触发轮胎印、打滑声和粒子效果。

• Enable Tire Smoke : Turns on/off tire smoke effects.

Enable Tire Smoke（启用轮胎烟雾）：打开或关闭轮胎烟雾效果。

• Enable Skidmark : Turns on/off skidmark effects.

Enable Skidmark（启用轮胎印）：打开或关闭轮胎印效果。

• Skidmark Material : Material of the skidmark mesh component.

Skidmark Material（轮胎印材质）：轮胎印网格组件使用的材质。

• Skidmark Max Marks : Maxiumum number of quads that can be generated by the skidmark system.

Skidmark Max Marks（轮胎印最大段数）：轮胎印系统能够生成的四边形最大数量。

• Skidmark Width : Width of the skidmark efffect in meters, try to make it close to the tire width.

Skidmark Width（轮胎印宽度）：轮胎印效果的宽度，单位为米，尽量接近胎宽。

• Skidmark Ground Offset : The height offset of the skidmark quads above the ground in meters.

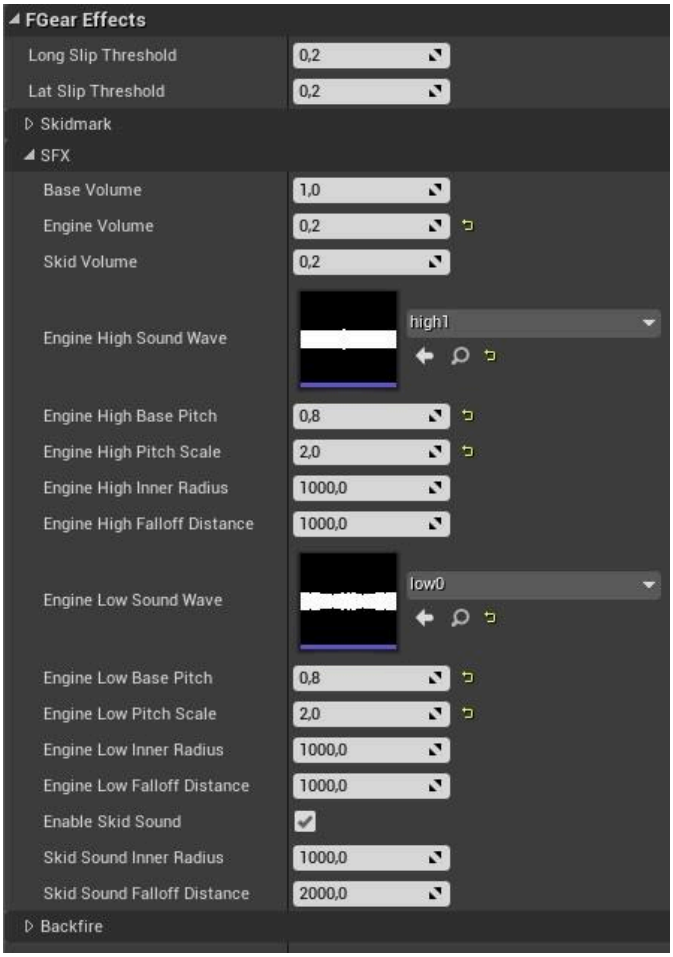
Skidmark Ground Offset（轮胎印离地偏移）：轮胎印四边形相对地面的高度偏移，单位为米。

• Skidmark Min Distance : Minumum distance between the quads of the skidmark mesh in meters.

Skidmark Min Distance（轮胎印最小间距）：轮胎印网格各四边形之间的最小距离，单位为米。

FGearEffects/SFX

FGearEffects / 音效



- **Base Volume : Multiplier to all sound effects.**

Base Volume（基础音量）：所有音效的乘数。

- **Engine Volume : Volume of the engine sound effect.**

Engine Volume（发动机音量）：发动机音效的音量。

- **Skid Volume : Volume of the skid sound effect.**

Skid Volume（打滑音量）：打滑音效的音量。

- **Engine High/Low Sound Wave : High/low pitch version of engine sound effect.**

Engine High/Low Sound Wave（发动机高/低音波形）：发动机音效的高音调和低音调版本。

- **Engine High/Low Base Pitch** : Minumum frequency of the engine(high/low) sound effect.**

Engine High/Low Base Pitch（发动机高/低音基础音高）**：发动机（高/低音）音效的最低频率。

- **Engine High/Low Pitch Scale** : This value is added to the base pitch value according to engines rpm value. At limit rpm final pitch = base + this**

Engine High/Low Pitch Scale（发动机高/低音音高缩放）**：根据发动机转速，把该值加到基础音高上。在限制转速时，最终音高 = 基础音高 + 该值。

- **Engine High/Low Inner Radius : Check unreals documents***

Engine High/Low Inner Radius（发动机高/低音内半径）：请查阅虚幻文档*。

- **Engine High/Low Falloff Distance : Check unreals documents***

Engine High/Low Falloff Distance（发动机高/低音衰减距离）：请查阅虚幻文档*。

- **Enable Skid Sound : Turn on/off skid sound effect.**

Enable Skid Sound（启用打滑声）：打开或关闭打滑音效。

- **Skid Sound Inner Radius : Check unreals documents***

Skid Sound Inner Radius（打滑声内半径）：请查阅虚幻文档*。

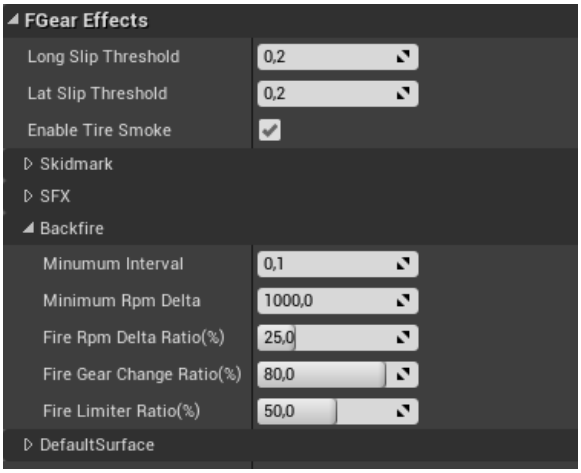
• **Skid Sound Falloff Distance : Check unreals documents***
Skid Sound Falloff Distance（打滑声衰减距离）：请查阅虚幻文档*。

* https://docs.unrealengine.com/en-US/Engine/Audio/DistanceModelAttenuation/index.html
* https://docs.unrealengine.com/en-US/Engine/Audio/DistanceModelAttenuation/index.html

****Set "ProjectSettings->Engine->Audio->Global Max Pitch Scale" to max value.**
**把 "Project Settings → Engine → Audio → Global Max Pitch Scale" 设为最大值。

FGearEffects/Backfire

FGearEffects / 回火



• **Minumum Interval : Minumum time between two backfire events(there is a %25 random variation).**
Minumum Interval（最小间隔）：两次回火事件之间的最短时间（另有 25% 的随机变化）。

• **Minumum Rpm Delta : If there is an rpm fall(rpm-delta/s) above this value, the event can be triggered.**
Minumum Rpm Delta（最小转速变化）：若转速下降（转速变化量/秒）超过该值，可以触发该事件。

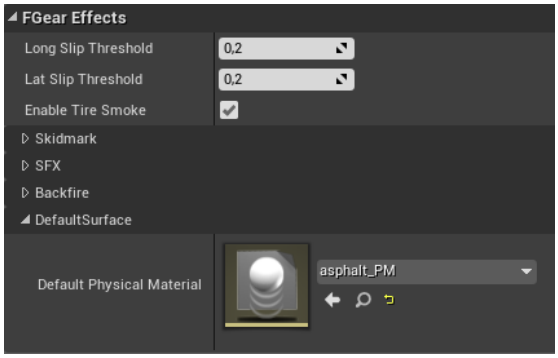
• **Fire Rpm Delta Ratio : Chance of trigger when rpm delta is above "Minumum Rpm Delta".**
Fire Rpm Delta Ratio（转速变化触发概率）：转速变化超过 "Minumum Rpm Delta" 时的触发概率。

• **Fire Gear Change Ratio : Chance of trigger on gear change.**
Fire Gear Change Ratio（换挡触发概率）：换挡时的触发概率。

• **Fire Limiter Ratio : Chance of trigger when rpm limiter is on.**
Fire Limiter Ratio（限速器触发概率）：转速限制器工作时的触发概率。

FGearEffects/DefaultSurface

FGearEffects / 默认表面



• **Default Physical Material : If the hit surface has no physical material assigned then we will use this material as default. If this is not set then no skidmark/smoke effect will be spawned.**
Default Physical Material（默认物理材质）：若命中表面没有指定物理材质，则使用该材质作为默认值。若此处也未设置，则不会生成轮胎印或烟雾效果。

FGearEffects/Events

FGearEffects / 事件



• On Brake State Changed : If braking state changes this event is triggered with the current brake "State" variable. You can use this event for a brake light effect.

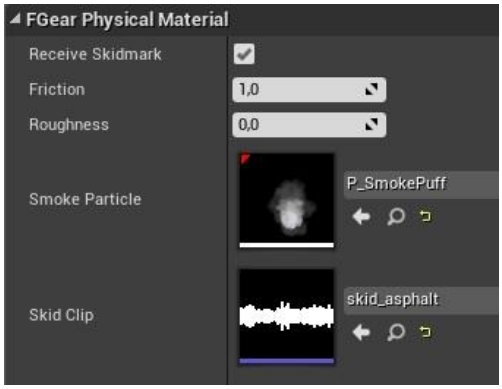
On Brake State Changed（制动状态改变）：制动状态变化时，以当前制动 "State" 变量触发该事件。可用于刹车灯效果。

• On Back Fire : This event is triggered according to the Backfire parameters explained above. You can use this event to trigger sound/visual effects.

On Back Fire（回火）：根据上文的回火参数触发该事件。可用于触发声音或视觉效果。

UFGearPhysicalMaterial : You can derive from this class and create your own surface types. This way you can alter tire friction and some visual effects.

UFGearPhysicalMaterial：可以从该类派生，创建自己的表面类型。这样就能改变轮胎摩擦以及部分视觉效果。



• Receive Skidmark : If set the skidmark mesh can be generated above this surface type.

Receive Skidmark（接受轮胎印）：若启用，可以在该表面类型上方生成轮胎印网格。

• Friction : Friction coefficient of the surface.

Friction（摩擦）：该表面的摩擦系数。

• Roughness : Not used currently.

Roughness（粗糙度）：当前未使用。

• Smoke Particle : The particle effect to spawn when skidding.

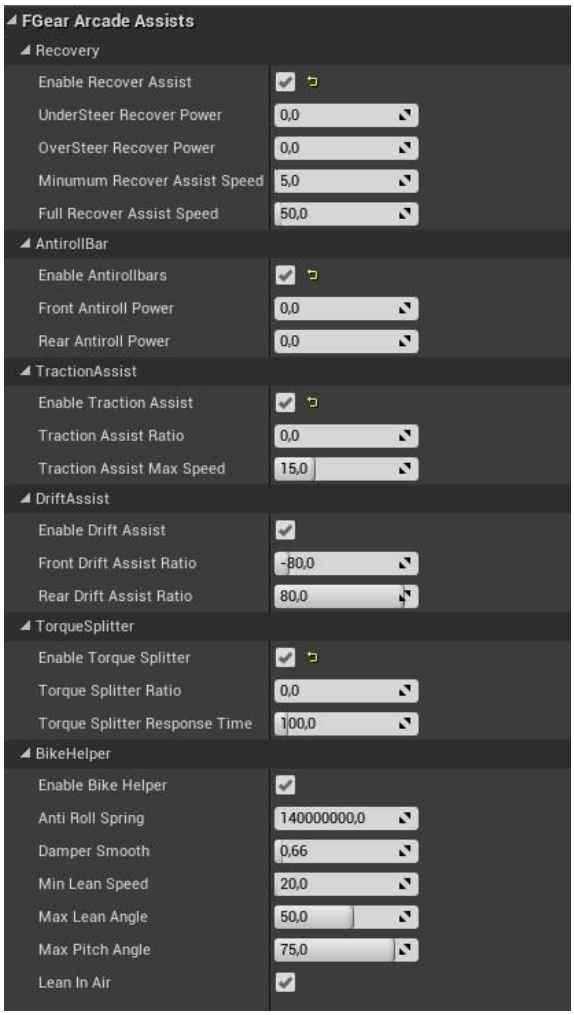
Smoke Particle（烟雾粒子）：打滑时生成的粒子效果。

• Skid Clip : The sound effect to play when skidding.

Skid Clip（打滑音效）：打滑时播放的音效。

UFGearArcadeAssists : Driving assists in the core module mostly uses a realistic approach but they are sometimes limited in efficiency. This component contains a couple of effective assists that can be used to get more controllable vehicle setups.

UFGearArcadeAssists：核心模块中的驾驶辅助大多采用较真实的做法，但有时效果有限。该组件包含若干更有效的辅助，用来得到更好控制的载具设定。



• Under Steer Recover : This one applies external torque to the vehicle body for avoiding understeer. This is a cheap way to battle understeering but combined with the over steer recover, you can easily get arcade style driving behaviour.

Under Steer Recover（转向不足恢复）：对车身施加外部扭矩，以避免转向不足。这是对抗转向不足的低成本办法；若再配合转向过度恢复，很容易得到街机风格的驾驶表现。

• Over Steer Recover : Applies external torque to the vehicle body for avoding over steer.

Over Steer Recover（转向过度恢复）：对车身施加外部扭矩，以避免转向过度。

• Minumum Recover Assist Speed : Below this speed(kmh) recover assists will have no effect.

Minumum Recover Assist Speed（恢复辅助最低速度）：低于该速度（km/h）时，恢复辅助不起作用。

• Full Recover Assist Speed : Below this speed(kmh) understeer/oversteer recover powers start to decrease linearly until minumum speed above.

Full Recover Assist Speed（恢复辅助全效力速度）：低于该速度（km/h）时，转向不足/转向过度恢复的力度开始线性下降，直到上面的最低速度。

• Anti Roll Bar(Front/Rear) : This is the legacy antirollbar implementation that applies external forces to avoid body roll during fast cornerings. Front antirollbar helps to reduce oversteering and rear antirollbar helps to reduce understeering. This only works for 4 wheel vehicles.

Anti Roll Bar（Front/Rear）（前/后防倾杆）：这是旧版防倾杆实现，在快速过弯时施加外力以减小车身侧倾。前防倾杆有助于减小转向过度，后防倾杆有助于减小转向不足。仅对四轮载具有效。

• Traction Assist : The tire model has a flaw that at low speeds the tires do not behave as expected and may loose traction. This assist helps to boost traction up to %200 at low speeds. Be aware that this modifies the lateral/longitudinal friction parameter of wheels.

Traction Assist（牵引力辅助）：轮胎模型有一个缺陷，低速时轮胎表现不符合预期，可能失去抓地。该辅助在低速时把抓地力最多提升到 200%。注意，它会修改车轮的横向/纵向摩擦参数。

• Traction Assist Max Speed : Traction assist is activated below this speed in kmh. The traction boost is scaled as the vehicle goes slower and reaches the maximum value when the vehicle stops.
Traction Assist Max Speed（牵引力辅助最高速度）：低于该速度（km/h）时启用牵引力辅助。车速越低，抓地提升越大，车辆停止时达到最大值。

• Drift Assist Front : As the vehicle goes sideways this value effects front wheels friction. Positive values result better grip as the oversteer angle increases and negative values result less grip as the oversteer angle increases. Giving negative values for both front and rear wheels will make the vehicle go sideways easily but you may need to use positive values for rear if the vehicle oversteers a lot. Tune it to get the desired behaviour. This only works for 4 wheel vehicles.
Drift Assist Front（前轮漂移辅助）：载具横向滑动时，该值影响前轮摩擦。正值会使转向过度角度越大抓地越好，负值会使转向过度角度越大抓地越差。前后轮都给负值时，载具更容易横滑；若转向过度很强，后轮可能需要使用正值。请调到想要的表现。仅对四轮载具有效。

• Drift Assist Rear : The same effect ratio for rear wheels. This only works for 4 wheel vehicles.
Drift Assist Rear（后轮漂移辅助）：后轮使用同一套效果系数。仅对四轮载具有效。

• Torque Splitter Ratio : If this is activated the torque share between front and rear axles are handled dynamically depending on their rotational speeds(still based on the the initial torque shares). Faster rotating axle will receive less torque and slower one will receive more even if the initial torque share is zero. This ratio is the percentage of torque share transfer. If this is %100 and your vehicle is rwd then in the extreme case all torque can be transferred to the front axle.
Torque Splitter Ratio（扭矩分配比）：若启用，前后轴之间的扭矩份额会按其转速动态分配（仍以初始扭矩份额为基础）。转得更快的车轴得到更少扭矩，转得更慢的车轴得到更多扭矩，即使初始扭矩份额为零。该比值是扭矩份额转移的百分比。若为 100% 且载具为后驱，极端情况下全部扭矩都可以转到前轴。

• Torque Splitter Response Time : Insant torque share transfer might be undesirable. This is the transfer time in miliseconds. The transfer will not happen instantly but will be interpolated in this duration.
Torque Splitter Response Time（扭矩分配响应时间）：瞬间转移扭矩份额可能不理想。这是转移时间，单位为毫秒。转移不会立刻完成，而是在该时长内插值。

• Antiroll Spring : This is the maximum torque that will be applied to keep the vehicle up.
Antiroll Spring（防倾弹簧）：为保持载具直立而施加的最大扭矩。

• Damper Smooth : Damper coefficient to the antiroll spring, lower values generate more damping torque.
Damper Smooth（阻尼平滑）：防倾弹簧的阻尼系数，数值越低产生的阻尼扭矩越大。

• Min Lean Speed : Below this speed the vehicle can not lean.
Min Lean Speed（最小侧倾速度）：低于该速度时载具不能侧倾。

• Max Lean Angle : Maximum angle the vehicle can lean.
Max Lean Angle（最大侧倾角）：载具能够侧倾的最大角度。

• Max Pitch Angle : Pitch angle is not limited while wheels have contact but when the vehicle is in air this limit kicks in and prevents flips.
Max Pitch Angle（最大俯仰角）：车轮着地时俯仰角不受限；载具在空中时该限制生效，防止翻滚。

• Lean in Air : If checked the vehicle can lean while flying.
Lean in Air（空中侧倾）：若勾选，载具在空中也可以侧倾。

• Enable Cruise Control : You can set a cruise speed and the vehicle will try to maintain that speed by using throttle and brakes.
Enable Cruise Control（启用巡航控制）：可以设定巡航速度，载具会用油门和制动尽量维持该速度。

• Cruise Speed : The target speed for the cruise control assist.
Cruise Speed（巡航速度）：巡航控制辅助的目标速度。

• PID Settings : With cruise control throttle and brakes are applied based on a PID controller.

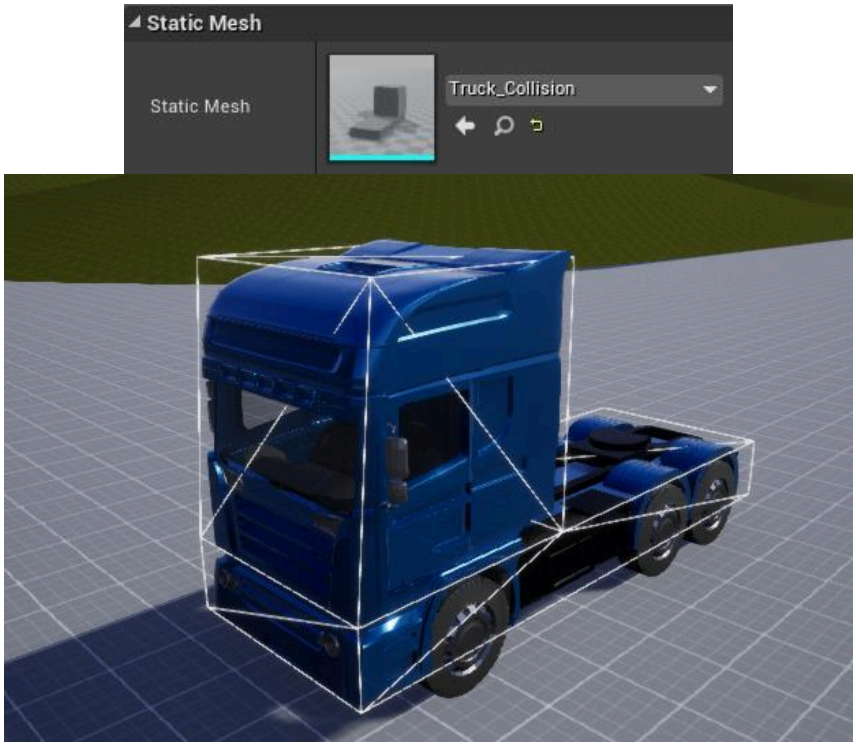
PID Settings（PID 设置）：启用巡航控制后，油门和制动依据 PID 控制器施加。

• setCruiseSpeed method : This is not a setting but a function that you can call to set the vehicles instant speed. This could be usefull for rolling starts. Check the DragRace map for a demonstration.

setCruiseSpeed 方法：这不是一项设置，而是一个可以调用的函数，用来设置载具的瞬时速度。可用于滚动发车。DragRace 地图中有演示。

UFGearCustomCollision : This optional collider component allows you to separate vehicles collision from inertial properties. For ex. you can use a separate mesh for an accurate representation of the vehicles body and use a box collider(with lower center of mass) in physical asset for the inertia. If this collider is not set the physical assets collider will be used for both collision and inertial properties but if you provide a custom mesh, the collision will be based on this mesh. You should untick CollisionResponse of physicsasset or both colliders will be active.

UFGearCustomCollision：这个可选碰撞组件可以把载具的碰撞与惯性属性分开。例如，可以用单独的网格精确表示车身，同时在物理资产里用一个盒体碰撞体（质心更低）表示惯性。若不设置该碰撞体，物理资产的碰撞体同时用于碰撞和惯性；若提供自定义网格，碰撞将基于该网格。应取消物理资产的 Collision Response，否则两套碰撞体都会生效。



Custom Collider Above, Physical Asset(Inertia) Below

上方为自定义碰撞体，下方为物理资产（惯性）

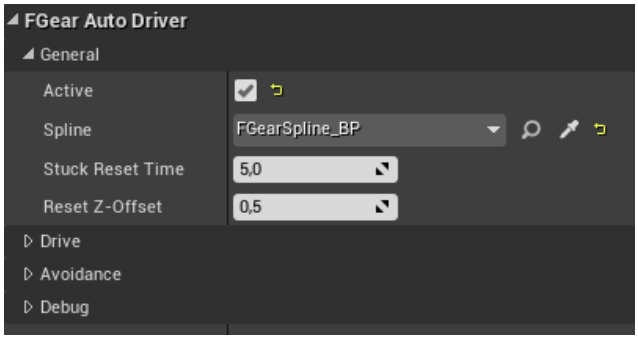


UFGearAutoDrive : Beginning with v1.8 this is no longer a component but an AI controller. So instead of adding

this as a component you need to possess a vehicle with this controller.
UFGearAutoDrive：从 v1.8 起，它不再是组件，而是一个 AI 控制器。因此不要把它作为组件添加，而要用该控制器 Possess 一辆载具。

This is an AI driver controller. You need to provide a USplineComponent that the ai can follow around a loop. AI tries to follow the spline, keep the speed according to curvature, try to avoid colliding other vehicles and reset if stuck. You can check the AITest map to see it in action.
这是一个 AI 驾驶控制器。需要提供一条 USplineComponent，供 AI 沿闭环跟随。AI 会尝试沿样条行驶，按曲率控制速度，尽量避免与其他载具碰撞，卡住时会重置。可以查看 AITest 地图中的实际效果。

The example controller of the sample project is in FGearAssets/OtherBpClasses/FGearAutoDrive_BP This controller looks for the first spline in the map at beginplay and assigns it.
示例工程中的示例控制器位于 FGearAssets/OtherBpClasses/FGearAutoDrive_BP。该控制器在 BeginPlay 时查找地图中的第一条样条并指定给自己。

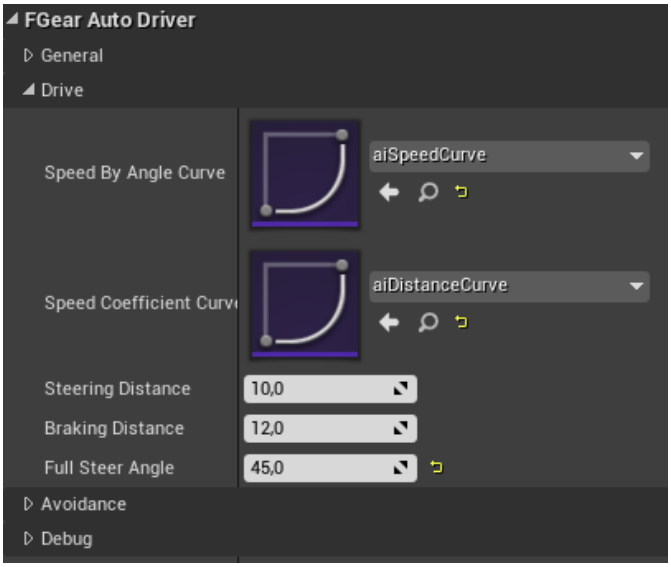


- **Active : Activates/deactivates the component.**
Active（启用）：启用或停用该组件。

- **Spline : FGearSpline actor to follow. FGearSpline has debug visualization features.**
Spline（样条）：要跟随的 FGearSpline Actor。FGearSpline 带有调试可视化。

- **Stuck Reset Time : If the vehicle is stuck it will reset on a position on spline after this amount of seconds.**
Stuck Reset Time（卡住重置时间）：若载具卡住，经过这么多秒后会重置到样条上的一个位置。

- **Reset Z-Offset : When the vehicle is reset on a spline position, it may collide with the ground so you can tweak this value to keep it above ground.**
Reset Z-Offset（重置 Z 偏移）：载具重置到样条位置时可能与地面穿插，可以调整该值使其保持在地面之上。



- **Speed By Angle Curve : For throttle and brake inputs the tangent angle of the spline and this curve is used. The x-axis of the curve is the rotation angle in degrees and the y-axis is the maximum speed value in kmh. AI tries to go as fast as the current speed limit acquired from this curve.**
Speed By Angle Curve（按角度限速曲线）：油门和制动输入使用样条切线角以及这条曲线。曲线 X 轴

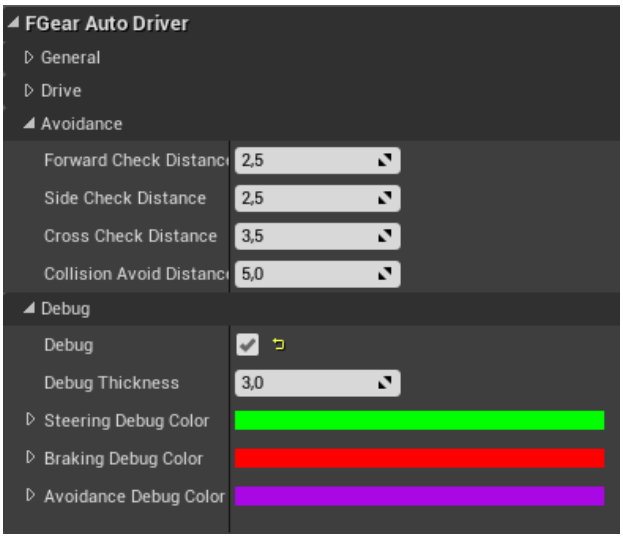
是旋转角度（度），Y 轴是最高速度（km/h）。AI 会尽量开到从该曲线取得的当前速度上限。

• Speed Coefficient Curve : The x-axis of this curve is speed in kmh and the y-axis is a coefficient that is multiplied with the steering and braking distance values. By using this curve you can make the AI look further as the speed increases. This also effects the collision avoidance checks.
Speed Coefficient Curve（速度系数曲线）：曲线 X 轴是速度（km/h），Y 轴是一个系数，会乘到转向距离和制动距离上。用这条曲线可以让 AI 在速度提高时看得更远。它也会影响避撞检测。

• Steering Distance : To calculate steering inputs we need to find a point on spline to check the curvature angle. We do this by moving on the spline this amount of distance in meters. As this value increases we steer earlier and as this value decreases we steer later.
Steering Distance（转向距离）：计算转向输入时，需要在样条上找一个点来检查曲率角。做法是沿样条前进这么多米。该值越大转向越早，越小转向越晚。

• Braking Distance : To calculate throttle/brake inputs we need to find a target speed. To find it we need to move on the spline this amount of distance in meters. We get the curvature angle and provide it to the speed by angle curve to find the speed limit. As this value increases we brake earlier and as this value decreases we brake later.
Braking Distance（制动距离）：计算油门/制动输入时，需要一个目标速度。为此要沿样条前进这么多米，取得曲率角，再交给按角度限速曲线得到速度上限。该值越大制动越早，越小制动越晚。

• Full Steer Angle : Steering input is a value between 0 and 1 and calculated as (curvature angle / full steer angle). So lowering this value makes the steering sharper.
Full Steer Angle（满转向角）：转向输入是 0 到 1 之间的值，按（曲率角 / 满转向角）计算。因此降低该值会使转向更急。



• Forward Check Distance : We make line traces to check for obstructions around the vehicle. This is the length of the line trace distance along the forward direction of the vehicle. Speed Coefficient Curve also effects the final value.
Forward Check Distance（前方检测距离）：用线段检测查看载具周围的障碍。这是沿载具前方的线段长度。速度系数曲线也会影响最终数值。

• Side Check Distance : This is the length of the line trace distance along the side direction of the vehicle.
Side Check Distance（侧向检测距离）：沿载具侧向的线段检测长度。

• Cross Check Distance : This is the length of the line trace distance along the corner direction of the vehicle.
Cross Check Distance（斜向检测距离）：沿载具对角方向的线段检测长度。

• Collision Avoid Distance : When a collision is detected, the vehicle will try to avoid by following a target position to the left or right of the collision target. This target position is this amount of meters away from the collision target.
Collision Avoid Distance（避撞距离）：检测到碰撞时，载具会去跟随碰撞目标左方或右方的一个目标位置。该目标位置距碰撞目标这么多米。

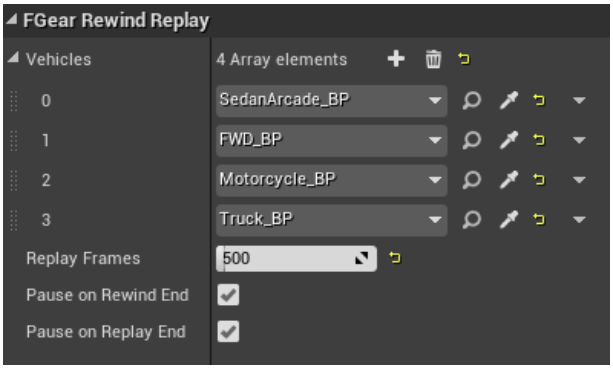
• **Debug : Shows debug visuals for the steering and braking points on the spline and also the line trace directions.**

Debug（调试）：显示样条上转向点和制动点的调试图形，以及线段检测方向。



AFGearRewindReplay : Place this actor in a map and assign the vehicles to be tracked. You can call some functions to rewind and replay the tracked vehicles. Check the RewindReplay.map for a demonstration. Warning : This feature is a framerate dependent solution and is not polished.

AFGearRewindReplay：把该 Actor 放到地图中，并指定要记录的载具。可以调用若干函数对已记录的载具倒带或回放。请查看 RewindReplay.map 中的演示。警告：该功能依赖帧率，而且尚未打磨完成。



• **Vehicles : The list of vehicles to be tracked.**

Vehicles（载具）：要记录的载具列表。

• **Replay Frames : The size of the history buffer for the vehicles. For ex. if you set this to 300 and if your game is locked to 60fps then you will store 5 seconds of history.**

Replay Frames（回放帧数）：载具历史缓冲的大小。例如设为 300，且游戏锁定在 60 帧，则会保存 5 秒历史。

• **Pause on Rewind End : The game will be paused when rewind reaches the start of history buffer.**

Pause on Rewind End（倒带到起点时暂停）：倒带到达历史缓冲起点时，游戏会暂停。

• **Pause on Replay End : The game will be paused when replay reaches the end of history buffer.**

Pause on Replay End（回放到终点时暂停）：回放到达历史缓冲终点时，游戏会暂停。